Lyapunov decay and a nonconvergent cold-start orbit with integral feedback

Abstract

We give a fixed analytic planar potential for an existing gradient system with integral feedback. Its published Lyapunov function is proper on the full phase space and decreases along every trajectory. Nevertheless, every sufficiently small positive displacement in one explicit zero-velocity, zero-memory family generates a bounded nonconvergent position orbit. Its full phase limit set is the circle of minimum equilibria. In the original time variable, the angle is asymptotic to $\sqrt{t}$, the velocity and memory norms to $1/(2\sqrt{t})$, and the energy to $1/(8t)$. The proof combines energy capture with a smooth asymptotic-phase projection and an exact positive quadratic memory coefficient. The equation, energy, invariant-manifold tools, and general nonconvergence phenomena are inherited; the result is an exact cold-start realization with these physical-time constants.

Keywords: Lyapunov function; integral feedback; bounded trajectory; stationary limit set; asymptotic phase

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1 The equation and the trajectory theorem

A decreasing Lyapunov function may control the phase without selecting one equilibrium. We examine this distinction for the continuous system

$$\dot{x} = v, \qquad \dot{v} = -v - 3g(x) - z - H(x)v, \qquad \dot{z} = g(x), \tag{1}$$

where $x, v, z \in \mathbb{R}^2$, $g = \nabla F$, and $H = \nabla^2 F$. We fix

$$F(x) = \left(1 + \frac{x_1}{2\sqrt{1 + \|x\|^2}}\right) \frac{(\|x\|^2 - 1)^2}{8}. \tag{2}$$

This is the PIDAO equation of [2], with gains $a = k_i = k_d = 1$ and $k_p = 3$. The gradient-PID family also appears in [1, equations (11)–(12)]. The cold initial condition is

$$x(0) = (0, 1 + \rho), \qquad v(0) = z(0) = 0. \tag{3}$$

In particular, $z(t) = \int_0^t g(x(s))\, ds$. These are the zero physical velocity and zero integral memory prescribed in [2, Supplementary Information, equation (C.8)]; no initial memory bias is inserted.

Put

$$w = v + z + g(x), \qquad E(x, v, z) = \frac{1}{2}\|w\|^2 + F(x) + \frac{1}{2}\|z\|^2. \tag{4}$$

This energy is the specialization of [2, Supplementary Information F]. The following theorem gives its capture property and the orbit behavior compatible with its decay. All limits refer to the original time in (1).

Theorem 1. *The function F in (2) is globally real analytic, nonnegative and coercive, and $F^{-1}(0) = \mathbb{S}^1$. The function E is proper on $\mathbb{R}^6$. Every initial phase generates a unique global forward solution of (1), and*

$$\frac{d}{dt}E(x(t), v(t), z(t)) = -\|w(t)\|^2 - \|g(x(t))\|^2. \tag{5}$$

For every cold start (3) with $0 < \rho \le 1/100$,

$$E(t) \le E(0) \le 6\rho^2, \qquad \sqrt{\tfrac{1}{2}} < \|x(t)\| < \sqrt{\tfrac{3}{2}} \quad (t \ge 0). \tag{6}$$

There is a constant $\rho_* \in (0, 1/100]$ such that every $0 < \rho < \rho_*$ has the following further properties:

$$\omega(x, v, z) = \{(x, 0, 0) : \|x\| = 1\}. \tag{7}$$

For the continuous angular lift with $\theta(0) = \pi/2$,

$$\frac{\theta(t)}{\sqrt{t}} \longrightarrow 1, \qquad \sqrt{t}\,\|v(t)\| \longrightarrow \frac{1}{2}, \qquad \sqrt{t}\,\|z(t)\| \longrightarrow \frac{1}{2}, \qquad tE(t) \longrightarrow \frac{1}{8}. \tag{8}$$

Consequently $x(t)$ does not converge and has infinite arc length, although $g(x(t))$, $v(t)$, $z(t)$ and $E(t)$ tend to zero.

Here ω is the set of limits of phase subsequences with times tending to infinity. The circle in (7) consists of equilibria; it is not a periodic orbit. The smallness threshold ρ_* is existential. The explicit interval $0 < \rho \le 1/100$ establishes (6), not all the assertions in (7)–(8).

Related nonconvergence results are classical. Cabot [3, Theorem 5.1] gives nonconvergence from zero initial memory for a different gradient-memory equation with convex flat minima. Bhat and Bernstein [4, Example 1.1] exhibit stationary-circle limit sets and identify additional conditions for point convergence. Horsin and Jendoubi [5, Theorem 1.1 and Remark 3] construct nonconvergent gradient trajectories and describe a heavy-ball transfer with a prescribed nonzero velocity. Their later work [6, Section 5] includes numerical analytic circle-minimum examples for nonlinear friction. The degenerate-damping discussion in [7, Proposition 4] is also pertinent. We do not claim a new general nonconvergence phenomenon. The issue here is the exact orbit of (1)–(3), with a fixed analytic potential and the constants in (8).

2 Lyapunov capture in the original phase

The multiplier in (2) is strictly between 1/2 and 3/2. Hence

$$F(x) \ge \frac{(\|x\|^2 - 1)^2}{16}. \tag{9}$$

This proves nonnegativity, coercivity and the stated minimum set. Global real analyticity follows from the positivity of $1 + \|x\|^2$. The minimum continuum shows, in particular, that F is not a convex function.

The change from (x, v, z) to (x, w, z) is a global smooth diffeomorphism, with inverse $v = w - z - g(x)$. Equation (1) becomes

$$\dot{x} = w - z - g(x), \qquad \dot{w} = -w - g(x), \qquad \dot{z} = g(x). \tag{10}$$

Differentiation of (4) gives

$$\dot{E} = \langle w, -w - g\rangle + \langle g, w - z - g\rangle + \langle z, g\rangle = -\|w\|^2 - \|g\|^2.$$

An energy sublevel bounds x by (9), and bounds w, z directly. It also bounds $g(x)$ and thus v. The sublevel is closed and compact in the original phase. Smooth local existence and uniqueness, followed by continuation on this compact set, proves global forward existence. Boundedness has therefore been established before any local invariant-manifold argument.

For the cold state write $R = 1 + \rho$ and $d = R^2 - 1$. Direct differentiation at $(0, R)$ gives

$$F = \frac{d^2}{8}, \qquad g = \left(\frac{d^2}{16\sqrt{1 + R^2}}, \frac{Rd}{2}\right), \qquad w = g.$$

Consequently

$$E(0) = \frac{(1 + R^2)d^2}{8} + \frac{d^4}{512(1 + R^2)}. \tag{11}$$

For $0 < \rho \le 1/100$, one has $R^2 < 2$ and $d < 3\rho$. Formula (11) gives $E(0) < 4\rho^2 < 6\rho^2$. From (9) and (5),

$$\big|\|x(t)\|^2 - 1\big| \le 4\sqrt{6}\,\rho < \frac{1}{2}, \qquad \|w(t)\|, \|z(t)\| \le \sqrt{12}\,\rho. \tag{12}$$

In particular, if $r = \|x\|$, then $|r - 1| + \|w\| + \|z\| \le C\rho$ for all future times, with one constant independent of t and ρ. This proves (6) and gives the uniform phase bound needed below.

3 An invariant graph and a smooth phase projection

Use the polar frame $e_r = (\cos\theta, \sin\theta)$ and $e_\theta = (-\sin\theta, \cos\theta)$. Set

$$n = r - 1, \qquad w = Ae_r + Be_\theta, \qquad z = Ce_r - pe_\theta.$$

Thus p is minus the tangential component of the integral memory. Write

$$G(n, \theta) = F_r(1 + n, \theta), \qquad T(n, \theta) = \frac{F_\theta(1 + n, \theta)}{1 + n}, \qquad \Omega = \frac{B + p - T}{1 + n}.$$

The exact equations on the annulus are

$$\begin{aligned} \dot\theta &= \Omega, & \dot n &= A - C - G,\\ \dot A &= -A - G + B\Omega, & \dot B &= -B - T - A\Omega,\\ \dot C &= G - p\Omega, & \dot p &= C\Omega - T. \end{aligned} \tag{13}$$

Define

$$\kappa(\theta) = 1 + \frac{\cos\theta}{2\sqrt{2}}. \tag{14}$$

Taylor expansion at the unit circle, uniformly in θ, gives

$$F(1+n,\theta) = \frac{\kappa(\theta)}{2}n^2 + O(n^3), \quad G = \kappa n + O(n^2), \quad T = \frac{\kappa'(\theta)}{2}n^2 + O(n^3). \tag{15}$$

The radial linear matrix for (n, A, C) is

$$L_\kappa = \begin{pmatrix} -\kappa & 1 & -1 \\ -\kappa & -1 & 0 \\ \kappa & 0 & 0 \end{pmatrix}, \qquad \det(\lambda I - L_\kappa) = \lambda^3 + (1+\kappa)\lambda^2 + 3\kappa\lambda + \kappa. \tag{16}$$

Since $\kappa > 0$ and $3\kappa(1+\kappa) > \kappa$, the cubic Hurwitz criterion implies that L_κ is Hurwitz. The fourth stable eigenvalue is -1, from B. The remaining linear equations are $\dot\theta = B + p$ and $\dot p = 0$. The equilibrium circle itself is not normally hyperbolic: there is a two-dimensional generalized zero space and a nontrivial zero Jordan block.

We give the required global angular construction instead of assuming a smooth projection from a formal local center expansion. All estimates in the next lemma are uniform in the angular variable. The coordinate scaling used in its proof never changes time.

Lemma 1. *There are $p_0 > 0$ and a C^6 invariant graph for* (13)*,*

$$(n, A, C, B) = h(\theta, p), \qquad \theta \in \mathbb{R}/(2\pi\mathbb{Z}), \quad |p| < p_0, \qquad h(\theta, 0) = h_p(\theta, 0) = 0. \tag{17}$$

On a neighborhood of the equilibrium circle there is a C^4 asymptotic-phase projection onto this graph, obtained by restriction of a compact normally attracting construction. It commutes with the flow. In a sufficiently small invariant neighborhood of the compact construction, an orbit tracks its projected orbit exponentially for all positive time. Whenever both orbits remain in the physical chart, this is a tracking statement for (1) *itself.*

Proof. Choose an auxiliary $\varepsilon > 0$ and set

$$(n, A, C, B) = \varepsilon(N, \mathsf{A}, \mathsf{C}, \mathsf{B}) = \varepsilon Y, \qquad p = \varepsilon s.$$

Regard s as a coordinate on $\mathbb{R}/(8\mathbb{Z})$, and choose a smooth periodic function $q(s)$ equal to s for $|s| \le 2$ and zero for $3 \le |s| \le 4$. In $|Y| < 3$ define

$$\widehat G = \frac{G(\varepsilon N, \theta)}{\varepsilon}, \qquad \widehat T = \frac{T(\varepsilon N, \theta)}{\varepsilon^2}, \qquad \tau = \frac{\mathsf{B} + q(s) - \varepsilon\widehat T}{1 + \varepsilon N}.$$

The quotients extend smoothly to $\varepsilon = 0$, by (15); their values are κN and $\kappa' N^2/2$. The periodic extension of the scaled equations is

$$\begin{aligned} \dot\theta &= \varepsilon\tau, & \dot s &= \varepsilon(\mathsf{C}\tau - \widehat T),\\ \dot N &= \mathsf{A} - \mathsf{C} - \widehat G, & \dot{\mathsf{A}} &= -\mathsf{A} - \widehat G + \varepsilon\mathsf{B}\tau,\\ \dot{\mathsf{C}} &= \widehat G - \varepsilon q(s)\tau, & \dot{\mathsf{B}} &= -\mathsf{B} - \varepsilon\widehat T - \varepsilon\mathsf{A}\tau. \end{aligned} \tag{18}$$

This is a definition outside $|s| < 2$, and introduces no derivative of q. Inside that strip it is exactly the scaled original field.

Let $u = (\theta, s) \in \mathbb{T}^2 = (\mathbb{R}/(2\pi\mathbb{Z})) \times (\mathbb{R}/(8\mathbb{Z}))$. Write (18) as

$$\dot u = \varepsilon f(u, Y, \varepsilon), \qquad \dot Y = L(\theta)Y + \varepsilon R(u, Y, \varepsilon), \qquad L = L_\kappa \oplus (-1). \tag{19}$$

Taylor's integral formula makes f, R smooth through $\varepsilon = 0$. Take a smooth cutoff $\chi(Y)$ which is one on $\|Y\| \le 1$ and zero on $\|Y\| \ge 2$. Multiply f and R by χ and leave $L(\theta)Y$ unchanged. For $|\varepsilon| < 1/8$ the denominators in (18) are positive on the support. The resulting field on $\mathbb{T}^2 \times \mathbb{R}^4$ is smooth and complete, has at most linear growth, and agrees with the literal scaled field for $|s| < 2$, $\|Y\| < 1$. On a neighborhood of the zero section it converges in every fixed finite C^j norm to

$$\dot u = 0, \qquad \dot Y = L(\theta)Y. \tag{20}$$

The compact torus $M_0 = \mathbb{T}^2 \times \{0\}$ is normally attracting for (20). To verify uniformity without choosing eigenvectors, let

$$P(\theta) = \int_0^\infty e^{tL(\theta)^T} e^{tL(\theta)}\, dt, \qquad L^T P + PL = -I. \tag{21}$$

The compact range of κ in (14) and (16) imply uniform exponential stability. The integral defines a smooth periodic positive-definite matrix with uniform upper and positive lower bounds. Thus, for some $K, \lambda_0 > 0$,

$$\|e^{tL(\theta)}\| \le K e^{-\lambda_0 t} \quad (t \ge 0).$$

The tangent flow of M_0 is the identity, so normal domination holds to every prescribed finite order.

Apply compact normally hyperbolic persistence at order six to a fixed-time map of the complete extended flow; see [8, Theorem 2.1 and Corollary 2.2]. Its hypotheses here are a compact boundaryless smooth manifold, an invariant splitting, uniform normal contraction, identity tangent dynamics, and an arbitrarily small C^6 perturbation near M_0. The resulting invariant torus is a C^6 graph

$$M_\varepsilon = \{Y = H_\varepsilon(u)\},$$

C^6-close to the zero section. Indeed base projection is a local diffeomorphism close to the identity and of degree one, hence a global diffeomorphism. Local uniqueness under persistence makes this time-map torus invariant under the continuous flow: a sufficiently short flow translate is another nearby invariant torus for the same time map. Local maximality also places the unchanged constant orbits $(\theta, s, Y) = (\theta, 0, 0)$ on M_ε.

We verify the regularity needed for the phase projection, which is additional to persistence. The reduced base field and its first derivative are $O(\varepsilon)$. Hence its tangent flow ϕ_t satisfies

$$\|D\phi_t|_{TM_\varepsilon}\| \le K_1 e^{c\varepsilon t}, \qquad m(D\phi_t|_{TM_\varepsilon}) \ge K_1^{-1} e^{-c\varepsilon t}, \tag{22}$$

where m denotes the minimum norm. In transverse coordinates $\eta = Y - H_\varepsilon(u)$, invariance gives

$$\dot\eta = [L(\theta) + O(\varepsilon)]\eta + O(\|\eta\|^2).$$

Since $\dot\theta = O(\varepsilon)$, differentiation of $\eta^T P(\theta)\eta$ gives $-\|\eta\|^2 + O(\varepsilon)\|\eta\|^2 + O(\|\eta\|^3)$. The invariant stable bundle therefore has a contraction rate $\lambda > 0$ independent of sufficiently small ε. Equivalently, the triangular variational system in (u, η) obtains its stable center component by integrating the decaying normal forcing against the inverse center fundamental matrix. This integral converges by (22) when $c\varepsilon < \lambda$. Thus

$$\|D\phi_t|_{E^s}\| \le K_2 e^{-\lambda t}.$$

Choose ε once so small that $7c\varepsilon < \lambda$. For $0 \le i \le 4$ this yields the center-bunching inequalities

$$\frac{\|D\phi_t|_{TM_\varepsilon}\|^i \|D\phi_t|_{E^s}\|}{m(D\phi_t|_{TM_\varepsilon})} \le K_i e^{[-\lambda + (i+1)c\varepsilon]t}. \tag{23}$$

The stable-foliation theorem with this regularity condition, in the form of [9, Corollary 2, equation (11)], gives a C^4 phase projection Π commuting with the flow. On a small invariant neighborhood,

$$d(\phi_t(y), \phi_t(\Pi y)) \le K_3 e^{-\lambda_1 t} d(y, \Pi y), \qquad t \ge 0, \tag{24}$$

for some $\lambda_1 > 0$. Such a neighborhood follows directly by taking a sufficiently small sublevel of $\eta^T P(\theta)\eta$ in the nonlinear inequality above, inside the foliated neighborhood. All constants may depend on the now fixed ε.

Undo the scaling in a smaller physical strip $|p| < p_0$. It yields (17). At each unchanged equilibrium the tangent plane of the graph is the generalized zero spectral space: the four other eigenvalues remain uniformly in the left half-plane, while the graph tangent dynamics is slow. In physical coordinates that space has $n = A = C = B = 0$. Hence $h_p(\theta, 0) = 0$. The restrictions of the graph and Π have the asserted regularity. The literal and auxiliary fields coincide wherever the cutoffs equal one, proving the last assertion of the lemma. ∎

4 The memory generated by a cold start

The graph in Lemma 1 is an actual invariant graph over the full angular circle. We now compute only its quadratic terms. Since $h = O(p^2)$, one has $\dot\theta = p + O(p^2)$ and $\dot p = O(p^3)$ there. Differentiating a quadratic graph along its reduced field contributes no quadratic term. If (n_2, A_2, C_2, B_2) are its quadratic coefficients, (13) gives

$$0 = A_2 - C_2 - \kappa n_2, \qquad 0 = -A_2 - \kappa n_2, \qquad 0 = \kappa n_2 - 1, \qquad 0 = -B_2.$$

It follows, uniformly in θ, that

$$n = \frac{p^2}{\kappa(\theta)} + O(p^3), \qquad A = -p^2 + O(p^3), \qquad C = -2p^2 + O(p^3), \qquad B = O(p^3). \tag{25}$$

The exact reduced dynamics therefore satisfies

$$\dot\theta = p + O(p^3), \qquad \dot p = -2p^3 + O(p^4). \tag{26}$$

The remainders are uniform Taylor bounds on the graph already constructed; they are not an assumption of invariance.

Let σ be the physical p coordinate of the phase projection Π. At an equilibrium,

$$D\sigma = dp. \tag{27}$$

Indeed the projection is the identity on the graph and its kernel is the stable space. All four stable directions have zero p component, as does the equilibrium tangent. The remaining generalized center direction is precisely the p direction. In particular, no linear B correction enters (27), although B does affect angular phase.

Lemma 2. *For the cold state y_ρ in* (3)*, the projected memory is*

$$\sigma(y_\rho) = \frac{7}{40\sqrt{2}}\rho^2 + O(\rho^3). \tag{28}$$

In particular it is positive for every sufficiently small positive ρ.

Proof. Fix an equilibrium angle and restrict to the radial variables $y = (n, A, C)$ with $B = p = 0$. Write the radial quadratic part of the phase function as

$$\sigma = p + Q_\theta(y) + O(\|y\|^3) \quad \text{on this radial slice.}$$

Equivariance of the phase projection and (26) imply that its derivative along the full flow has no term of degree two on this slice. By (13) and (15), the quadratic term of $\dot p$ is $-\kappa' n^2/2$. Thus

$$DQ_\theta(y)L_\kappa y = \frac{\kappa'(\theta)}{2}n^2. \tag{29}$$

The first-order angular velocity vanishes on this radial slice, so no angular derivative of Q_θ contributes at degree two. Since L_κ is Hurwitz, integrating (29) along its exponentially decaying linear orbit gives

$$Q_\theta(y) = -\frac{\kappa'(\theta)}{2} \int_0^\infty \left[e_1^T e^{tL_\kappa} y\right]^2 dt. \tag{30}$$

This is an identity for a Taylor coefficient of the smooth invariant phase map. No finite-time trajectory approximation is extended to infinite time.

At $\theta_0 = \pi/2$, $\kappa = 1$ and $\kappa' = -1/(2\sqrt{2})$. The literal cold state satisfies

$$(n, A, C) = \rho(1, 1, 0) + O(\rho^2), \qquad B = O(\rho^2), \qquad p = 0.$$

The first-order radial vector is $(1, 1, 0)$ because $w(0) = g(x(0))$, although $v(0) = 0$. Equation (27) shows that the second-order B component makes no contribution to (28); its quadratic cross terms are of order at least three.

Let $M = L_1$ and

$$S = \begin{pmatrix} 2/5 & 1/10 & 0 \\ 1/10 & 1/10 & -1/10 \\ 0 & -1/10 & 3/10 \end{pmatrix}. \tag{31}$$

Direct multiplication gives $M^T S + SM = -\operatorname{diag}(1, 0, 0)$. Therefore

$$\int_0^\infty \left[e_1^T e^{tM}(1, 1, 0)^T\right]^2 dt = (1, 1, 0) S (1, 1, 0)^T = \frac{7}{10}.$$

Combining this equality with (30) proves (28). The C^4 regularity of the phase map makes the cubic remainder finite for the fixed auxiliary ε. This proves positivity for an entire sufficiently small interval of positive ρ. ∎

5 All-future comparison and the limit set

We first fix the neighborhoods, so that the reduced calculation applies to the actual cold orbit for all time. Shrink p_0 in Lemma 1 until the uniform remainder in (26) gives

$$-3p^3 \le \dot{p} \le -p^3 \qquad (0 < p < p_0), \tag{32}$$

and the graph lies in the region where the auxiliary cutoffs equal one. There are only finitely many further smallness requirements on ρ. Choose one $\rho_* > 0$, no larger than $1/100$, such that, for every $0 < \rho < \rho_*$, the scaled cold point belongs to the invariant foliated neighborhood, its projected point has $0 < \sigma(y_\rho) < p_0/2$, and the two angular phases start sufficiently close for their tracked difference to have a continuous small lift. These requirements follow from Lemma 2, smoothness of Π, and continuity at the equilibrium.

In addition, the already proved estimate (12) lets us decrease the same ρ_* so that the entire literal orbit satisfies

$$|p(t)/\varepsilon| < 1, \qquad \|(n(t), A(t), C(t), B(t))/\varepsilon\| < \frac{1}{2} \quad (t \geq 0).$$

This requirement uses only the original energy. It does not assume tracking, an invariant graph approximation, or an asymptotic limit. Along this entire orbit the original and extended vector fields agree. Uniqueness identifies their solutions for all future time.

The projected orbit also remains in the physical part of the graph. Its p coordinate is positive and decreasing by (32). It cannot reach zero in finite time by uniqueness, and its limit is zero: a positive limit would make its derivative uniformly negative. The angular coordinate ranges over a full circle, so it cannot leave the angular chart. Formula (25) keeps the other graph coordinates inside the cutoff-equality region. We may therefore use (24), transported through the fixed smooth coordinate changes, between the actual cold orbit and this actual graph orbit for every $t \geq 0$.

We prove the limits first on the graph. Equations (26) and (32) imply

$$\frac{d}{dt} p^{-2} = 4 + O(p) \longrightarrow 4.$$

After integration,

$$p(t)\sqrt{t} \longrightarrow \frac{1}{2}. \tag{33}$$

Also $\dot{\theta} = p + O(p^3)$. For each fixed $\delta > 0$, (33) bounds $p(t)$ between $(1-\delta)/(2\sqrt{t})$ and $(1+\delta)/(2\sqrt{t})$ for sufficiently large t. Integrating these bounds shows

$$\frac{1}{\sqrt{t}} \int_0^t p(s)\, ds \longrightarrow 1.$$

The tail integral of p^3 is finite, again by (33). Hence $\theta(t)/\sqrt{t} \to 1$ for any graph lift.

On the graph, (25) gives $r = 1 + O(p^2)$ and

$$z = Ce_r - pe_\theta = -pe_\theta + O(p^2), \qquad w = O(p^2), \qquad g = O(p^2), \qquad F = O(p^4).$$

Since $v = w - z - g$, it follows that

$$\frac{\|v\|}{p} \longrightarrow 1, \qquad \frac{\|z\|}{p} \longrightarrow 1, \qquad \frac{E}{p^2} \longrightarrow \frac{1}{2}. \tag{34}$$

These formulas and (33) prove the remaining constants in (8) on the graph.

Exponential tracking transfers these limits to the original cold orbit. All coordinate changes have bounded derivatives on the captured neighborhood. The energy is smooth there, so the energy difference is also exponentially small. The angular distance has a continuous small lift tending to zero; an alternative choice of lift only changes it by a fixed multiple of 2π. Thus the specified literal lift with $\theta(0) = \pi/2$ has the same normalized limit. In particular it tends to positive infinity.

Every phase limit point has $r = 1$ and $v = z = 0$. Conversely, for each prescribed angle ϑ modulo 2π, the continuous literal lift reaches $\vartheta + 2\pi j$ for every sufficiently large integer j. The corresponding times tend to infinity. Along them $r \to 1$ and $v, z \to 0$. This proves the equality in (7). It also proves position nonconvergence. Finally (8) yields $\|v(t)\| \geq 1/(4\sqrt{t})$ for all sufficiently large t, proving infinite arc length and completing Theorem 1.

6 Interpretation and source scope

The Lyapunov function (4) provides the essential all-future capture estimate. Its dissipation observes w and g, whereas the motion along the invariant graph has $v \sim pe_\theta$ and $z \sim -pe_\theta$. Thus the leading tangential velocity is larger than the directly dissipated quantities. This explains why the same energy can tend to zero while the position continues around the minimum circle. The argument establishes the actual orbit and its time laws, rather than inferring nonconvergence from a missing energy estimate.

Theorem 3 of [2] displays limits of the gradient and of $v + (k_i/a)z$, rather than point convergence of x. Our example is consistent with those displayed conclusions. That theorem also uses the literal phrase "bounded objective function." The coercive potential (2) is not globally bounded above. We therefore use the source energy as an identity and prove capture directly, without silently replacing a printed hypothesis. The potential here is analytic and nonconvex; the convex and strongly convex theorems in that paper are different statements.

Analyticity of an energy does not by itself verify an angle condition linking dissipation to the full phase speed. For example, the quasi-gradient convergence theorem in [7, Definition 3 and Theorem 6] includes a directional inequality and equality of the energy-critical and stationary sets. We have not used such a theorem without its hypotheses. Nor is the $t^{-1/2}$ law presented as a new general mechanism: it comes from the cubic reduced damping in (26). The specific coefficient generated by the exact cold start is what connects that reduction to (1).

The invariant-manifold argument uses standard compact persistence and smooth stable foliations [8, 9]. The auxiliary torus is a proof construction. Neither a normally hyperbolic equilibrium circle nor a new invariant torus of the physical system is asserted. In particular, the potential and all gains stay fixed as $\rho \downarrow 0$, and physical time is never rescaled.

The literature comparison is limited to the inspected versions. The exact equation and energy in [2], Cabot's author version [3], the conference example in [4], and the author versions in [5, 6] supply explicit ancestry. Full theorem texts for several recent distributed, time-dependent and fractional PID followups, some classical journal-final versions, and a related 2015 corrigendum were not available in the comparison. Their advertised models

differ, but unread sections are not exclusions. No firstness or exhaustive priority claim is made. The accompanying technical note records these version and coverage limits more precisely.

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